ENGINEERING PORTFOLIO

ANDREW MARION

MECHANICAL ENGINEERING · AEROSPACE

Junior in Mechanical Engineering with an Aerospace minor at Kennesaw State University. My work splits between hands-on hardware (rocket propulsion, autonomous aircraft, ground support equipment) and CFD research validated against wind-tunnel and experimental data. Currently a UAV Flight Control Intern at NASA Glenn Research Center and Vice President of the Kennesaw Rocketry Team.



3.81
GPA · KSU

May '27
GRADUATION

4
SCITECH 2027 PAPERS ACCEPTED

140+
CFD SIMULATIONS RUN

CONTACT

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researchgate.net/profile/Andrew-Marion

EDUCATION

Kennesaw State University
B.S. Mechanical Engineering, Aerospace Engineering Minor
Anticipated May 2027 · U.S. Citizen

COURSEWORK

Aircraft Design, CAE, Aerodynamics, Fluid Dynamics, Heat Transfer

CONTENTS

Internships · Build Projects · Research · Publications · Skills

01 INDUSTRY INTERNSHIPS

NASA GLENN RESEARCH CENTER — CLEVELAND, OH AUG – DEC 2026

INTERNSHIP

UAV FLIGHT CONTROL INTERN

Diagnostics and Controls Branch · Distributed electric propulsion UAV based on NASA's SUSAN Electrofan



- Implementing custom MATLAB/Simulink flight control algorithms on Pixhawk flight computers running PX4 for a 12 ft wingspan distributed electric propulsion UAV with 16 electric fans and thrust vectoring
- Developing control reallocation algorithms in MATLAB that redistribute thrust commands across the remaining fans after a fan or thrust-vectoring failure, tested across 8,000 fan-failure cases with Monte Carlo simulation
- Modeling flight dynamics and powertrain in MATLAB/Simulink and X-Plane, and performing system identification on flight test data with Kalman filtering and airspeed, IMU, and GPS sensor fusion
- Integrating algorithms into PX4 in a Linux environment; invited to return in spring 2027 to continue the project

MATLAB/Simulink PX4 Pixhawk Kalman Filters System ID X-Plane



TEXTRON GSE — CARTERSVILLE, GA JUN – AUG 2026

INTERNSHIP

MECHANICAL DESIGN ENGINEER INTERN

Ground support equipment electrification

- Designed assemblies in Siemens NX and produced build-ready drawings, including a hinged door for the MX4 electric baggage tractor and access doors for the 660 belt loader refresh, with hands-on sheet metal design and prototyping
- Applied GD&T and tolerance stack-up analysis alongside DFA/DFM reviews to validate manufacturability before production release
- Ran fit checks and physical prototype testing, including parking-brake sensor location testing and access-door motion verification

Siemens NX GD&T Sheet Metal DFA/DFM

KSU OFFICE OF RESEARCH — MARIETTA, GA FEB – MAY 2026

INTERNSHIP

RESEARCH DEVELOPMENT INTERN

Funding and grant support for KSU faculty

- Conducted funding searches and budget analyses to identify external grant opportunities for KSU faculty
- Designed and built web pages that communicated funding opportunities and research resources to faculty
- Built NIH R15 and R16 grant resources, including checklists, to help faculty prepare proposals

Funding Search Budget Analysis Grant Resources SharePoint Web Design

02 BUILD PROJECTS



AERIAL ROBOTICS COMPETITION TEAM SEP 2023 – APR 2025

TEAM LEAD

POWERED AUTONOMOUS DELIVERY AIRCRAFT (PADA)

SAE AERO Design — Team Lead, 20+ members

- Led the full design, manufacturing, and flight-test lifecycle of a carrier-launched delivery aircraft, from CAD through fabrication and integration
- Manufactured a carbon fiber mold and performed a wet layup to construct the fuselage
- Built and flew a custom flying wing that detached mid-flight from a larger carrier aircraft and autonomously navigated to landing coordinates it detected in flight
- Configured and tuned a Matek H743 flight computer for stable flight on an inherently unstable airframe



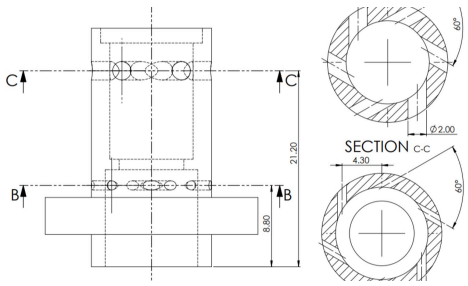
KENNESAW ROCKETRY TEAM DEC 2025 – PRESENT

OPERATIONS / CFD

IREC COMPETITION ROCKET

Vice President — Team Operations Lead

- Led team operations for the Intercollegiate Rocket Engineering Competition build across Rocket Design, Avionics, and Payload subteams
- Optimized rocket body and camera mount geometry against drag through high-fidelity CFD analysis
- Assisted with fiberglass construction and resolved fin-to-airframe gap issues through iterative SolidWorks corrections
- 3D printed the camera mount and payload components; standardized build documentation



KENNESAW ROCKETRY TEAM DEC 2025 – PRESENT

LIQUID PROPULSION

BIPROPELLANT ROCKET ENGINE

Vice President; previously Liquid Propulsion Lead

- Designing and sizing an N_2O / ethanol bipropellant engine targeting 250 lbf thrust at an O/F ratio of 4.8: thrust equations, injector pressure drop, chamber pressure, and mass-flow calculations
- Designing an injector cold-flow test stand around a pressurized nitrogen supply (~150–220 psi)
- Reviewed and refined a coaxial swirl injector assembly through SolidWorks design checks ahead of fabrication

02 BUILD PROJECTS (CONT.)



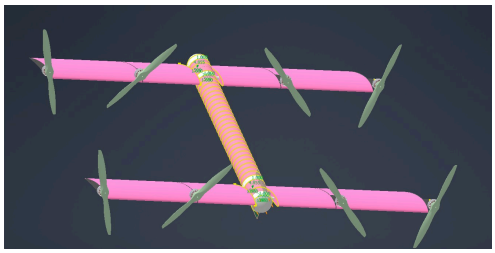
AIRCRAFT DESIGN AND PERFORMANCE — TEAM OF 4 SPRING 2026

CONCEPTUAL DESIGN

SHIP-LAUNCHED MULTI-DOMAIN ISR UAV

ENGR 3802 · Advisor: Dr. Gaurav Sharma · Responsible for airfoil selection, CAD/CFD, and final sizing

- Designed a ship-launched, tailless flying-wing UAV for air and underwater ISR: 15.4 ft span, 40° leading-edge sweep, 2,050 lb takeoff weight, and a single internal PW610 turbofan
- Met every requirement with margin: 1,060 NMi range against 1,000, 47 min loiter against 30, and 89 kt stall against a 97 kt limit, with a cruise L/D of 13.5 at Mach 0.7
- Developed the dual-domain transition: deep-stall water landing at about 10° pitch, flooded wing-root tanks as ballast, and retractable wingtip thrusters for underwater transit
- Chose a hybrid CFRP and aluminum 5083 structure with Inconel in the exhaust region and EPDM seals



DARPA LIFT CHALLENGE — TEAM OF 6 STUDENTS DESIGN PROJECT

CONCEPTUAL DESIGN

TANDEM-WING VTOL UAV FOR THE DARPA LIFT CHALLENGE

Aerodynamic and planform design lead

- Led the aerodynamic and planform design of a tandem-wing VTOL UAV concept for a challenge calling for a drone under 55 lb that lifts at least 110 lb, a 4:1 payload-to-weight ratio
- Designed a blown-wing configuration with eight propellers across two tandem wings, using propeller slipstream over the wings to add lift
- Built a fuselage and wing sections from foam core, with fiberglass rebar as the structural support
- Completed as a student design project, not as an entry in the competition



DYNAMICS — KENNESAW STATE UNIVERSITY FALL 2024

COURSE PROJECT

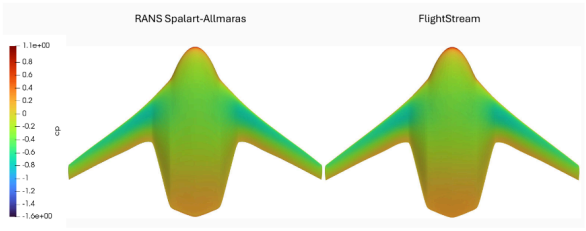
PUMPKIN LAUNCH TREBUCHET

Design, build, and launch

- Designed and built a 12 ft trebuchet with a 6 ft arm for Kennesaw State's annual pumpkin launch challenge, capable of throwing 15 lb pumpkins
- Performed engineering analysis from first principles and dynamics to size the arm, sling, and counterweight
- Built it from primarily wood, using buckets filled with poured cement as a 200 lb counterweight

03 CFD & AERODYNAMICS RESEARCH

KSU Advanced Aerodynamics and Propulsion Lab: simulation work validated against wind-tunnel and experimental data, published at AIAA conferences.

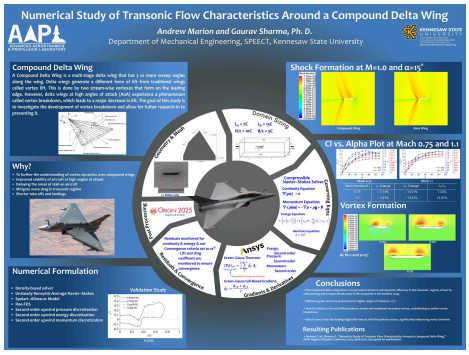


JAN 2026 – PRESENT 1ST AUTHOR ×2

BLENDED WING BODY VS. RANS CFD

Siemens FlightStream benchmarking

- Benchmarking FlightStream against RANS-based CFD for a blended wing body (BlendedNet++ dataset, Geometry 227) in collaboration with Siemens
- Assessing FlightStream's potential to speed up early-stage aircraft concept simulation
- Two papers accepted as first-author work at AIAA SciTech 2027

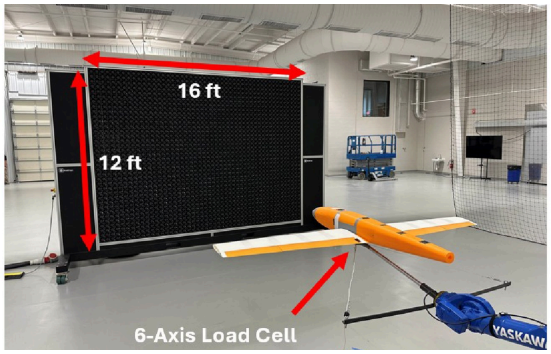


SEP 2023 – PRESENT 1ST AUTHOR

COMPOUND DELTA WING CFD

Vortex lift & transonic performance

- Executed 140+ CFD simulations in ANSYS Fluent across velocities and angles of attack, improving lift-to-drag efficiency from 7.24% to 13.57% in the transonic regime
- Investigated vortex lift modeling through trailing-edge boundary condition treatment
- First-authored paper at the 2025 AIAA Region II Student Conference

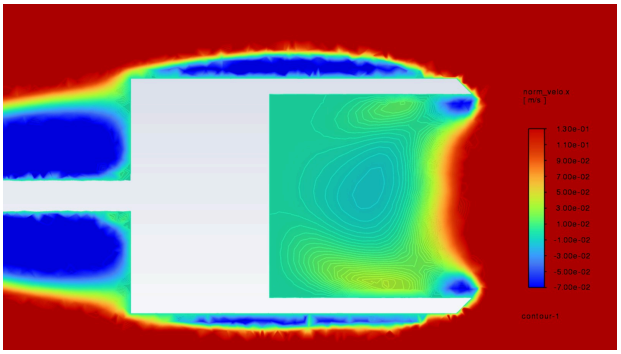


MAR 2025 – PRESENT CO-AUTHOR

CONTROL SURFACE VALIDATION — "BIRD" UAV

KSU × Oklahoma State S2FAR Lab

- Validated FlightStream aileron-deflection load predictions against high-fidelity CFD and WindShape wind-tunnel / load-cell testing on a Group-1 fixed-wing UAV
- Created aerodynamic models informing initial flight conditions for closed-loop roll-axis control validation on a Pixhawk Cube Orange+
- Co-authored findings at AIAA SciTech 2026



IN PROGRESS DESIGN PHASE

FORWARD-FACING CAVITY PIV REPLICATION

URANS CFD vs. experimental PIV

- Running URANS CFD to replicate a forward-facing cylindrical cavity PIV experiment, working through timestep selection and Rossiter mode analysis
- Coordinating validation of vorticity and velocity distribution against experiment

04 PUBLICATIONS & PRESENTATIONS

JAN 2027 · ACCEPTED
Aerodynamic Characterization of a Blended Wing Body Using a Surface Vorticity Solver
Marion, A., Sharma, G., Sirak, E., and Shahjahan, S. · AIAA SciTech 2027 Forum

JAN 2027 · ACCEPTED
Modeling Vortex-Dominated Flow over a Delta Wing Using a Surface Vorticity Solver
Marion, A., Sharma, G., Sirak, E., and Shahjahan, S. · AIAA SciTech 2027 Forum

JAN 2027 · ACCEPTED
Aerodynamic Analysis of a Blended Wing Body Using the Lattice Boltzmann Approach
Mehar, P., Marion, A., Sharma, G., Muller, M., and Gaedtke, M. · AIAA SciTech 2027 Forum

JAN 2027 · ACCEPTED
Cross-Validation of a Mid-Fidelity Fixed-Wing Aerodynamic Model for Control Law Development with Flight Testing
Atkinson, Z., Comer, A., Marion, A., and Sharma, G. · AIAA SciTech 2027 Forum

JAN 2026 · CONFERENCE PAPER
Multi-Fidelity Validation of Control Surface Deflection Predictions Using FlightStream
Atkinson, Z., Comer, A., Marion, A., Sharma, G., Shahjahan, S., and Wilson, T. · AIAA SciTech 2026 Forum · doi.org/10.2514/6.2026-2810

MAY 2025 · INVITED TALK
Study of Compound Delta Wings and Kennesaw State Undergraduate Research Experience
Marion, A. · Presented to the President's Community Advisory Board

APR 2025 · CONFERENCE PAPER
Numerical Study of Transonic Flow Characteristics Around a Compound Delta Wing
Marion, A., and Sharma, G. · 2025 AIAA Region II Student Conference · doi.org/10.2514/6.2025-97857

MAY 2024 · POSTER
Numerical Analysis of Vortex Breakdown Over a Compound Delta Wing
Marion, A., and Sharma, G. · Symposium of Student Scholars

MAR 2026 · FEATURE STORY
Engineering Student Uses Computer Simulations to Shape the Future of High-Speed Flights
Kennesaw State University News

05 TECHNICAL SKILLS

CFD SOFTWARE
ANSYS Fluent, FlightStream, SolidWorks Flow Simulation, OpenVSP, ParaView

CAD SOFTWARE
SolidWorks, Siemens NX, AutoCAD, ANSYS SpaceClaim

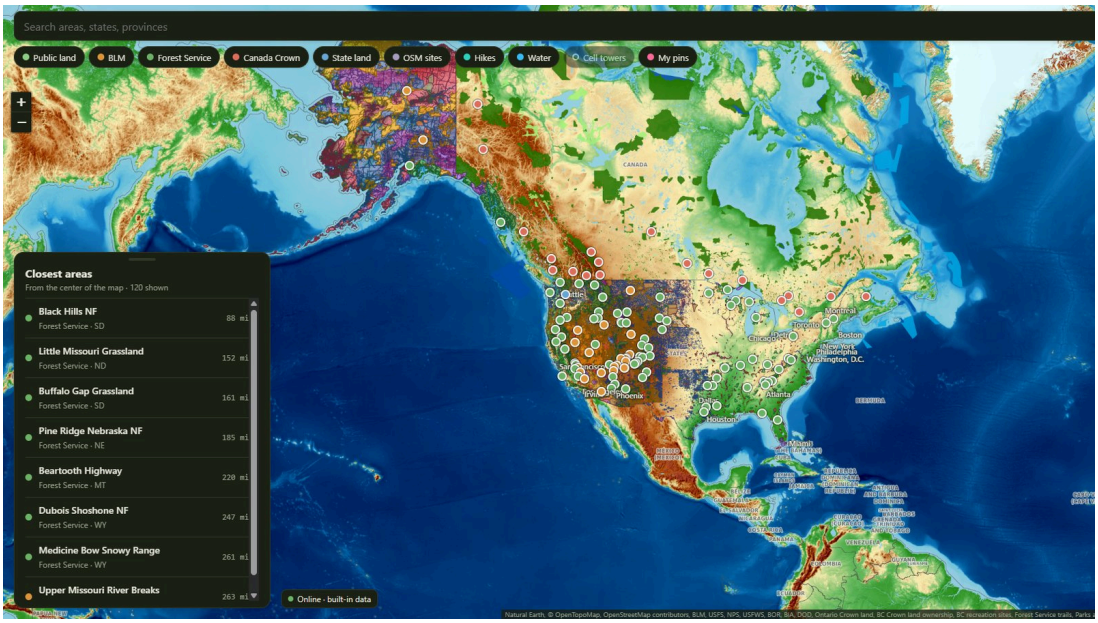
PROGRAMMING / AUTOPILOT
MATLAB/Simulink, Python, R, LaTeX, PX4, ArduPilot, Pixhawk

FABRICATION & SHOP
Sheet metal prototyping, 3D printing, laser cutting, composite layup, lathe/mill, CNC machining, foam cutting

DESIGN PRACTICE
GD&T, tolerance stack-up, DFA/DFM reviews, fit checks, build documentation

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06 GIS MAPPING



PERSONAL PROJECTS OCT 2026 – PRESENT

GIS / MAPPING

DISPERSED CAMP MAP AND 3D HIKING TERRAIN MAP

Web maps built from open geospatial data, terrain elevation data, and GPS metadata

- Built the Dispersed Camp Map, a single-file Leaflet web app that overlays color-coded public land (BLM, Forest Service, parks, state, tribal, military) from BLM Surface Management Agency data, so tapping any spot shows who manages it
- Integrated multiple GIS data sources: US Forest Service trails and road-use maps, Ontario and BC Crown land, wildfire perimeters, and OpenStreetMap campsites, water points, and cell towers
- Added offline regions with USGS topo tiles, GPS track recording with GPX export, and saved pins with photos, ratings, and signal notes, all stored on the device; packaged as an installable progressive web app that launches with no signal
- Built an interactive 3D terrain map of my hiking photos in MapLibre GL JS: a Python script reads the GPS coordinates, elevation, and timestamp from each photo's EXIF metadata to place the pins, over satellite imagery, open elevation data, and hillshading

Leaflet | MapLibre GL JS | Python | EXIF / GPS | Terrain / DEM | PWA

07 HIKING



From the Appalachian Trail in Georgia to Exit Glacier in Alaska. An interactive map of all 19 stops is on my website.